

Recruitment System

1. Introduction

This project is a recruitment system database designed to manage job applications, companies, departments, and users. The system allows users to apply for jobs, companies to post jobs, and departments to organize job roles.

This project addresses a real-world complex computing problem by managing multiple entities and their relationships efficiently.

2. Complex Computing Problem

The recruitment system involves:

- Multiple entities (Users, Jobs, Applications, Resumes)
- Relationships between entities
- Constraints like unique applications
- Real-world scenario (job hiring process)

This makes it a complex computing problem because:

- Data integrity is maintained
- Multiple relationships exist
- Constraints prevent invalid data

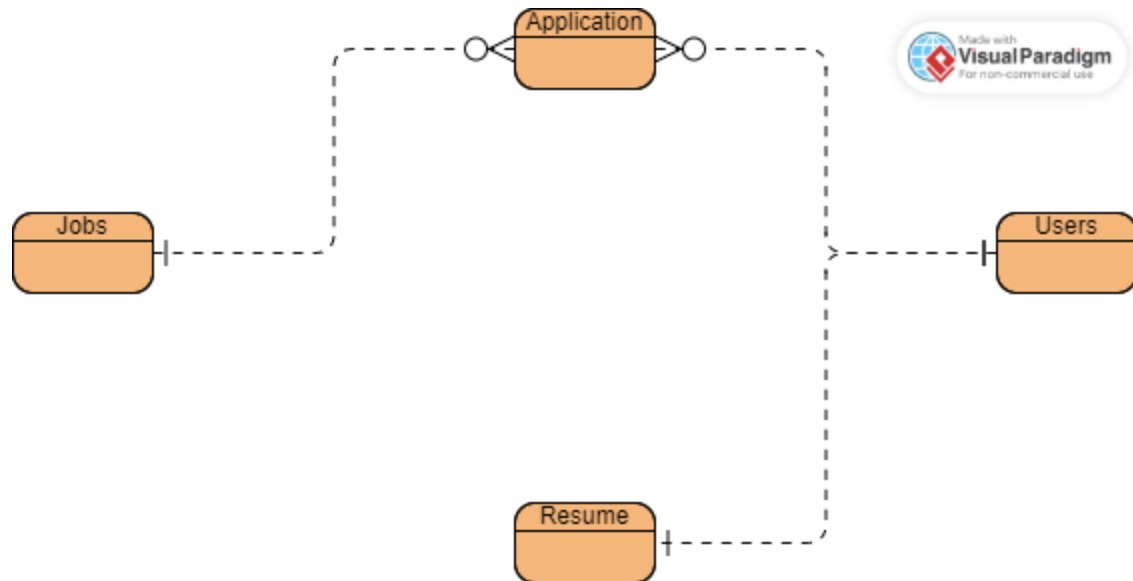
3. ERD

Entities:

- Users
- Resumes
- Jobs
- Application

Relationships:

- One User $\rightarrow$ One Resume
- One User $\rightarrow$ Many Applications
- One Job $\rightarrow$ Many Applications



4. Schema Design

Tables:

Users

- user_id (PK)
- user_name
- user_email
- user_phone_no

Resumes

- resume_id (UK)
- user_id (FK)

Jobs

- job_id (PK)
- department_id
- company_id

Application

- app_id (PK)
- user_id (FK)
- job_id (FK)

5. Normalization

The database is normalized up to 3NF:

1NF:

- No repeating groups
- Atomic values

2NF:

- No partial dependency
- All attributes depend on primary key

3NF:

- No transitive dependency
- All non-key attributes depend only on primary key

6. Constraints Used

- Primary Key → Unique identification
- Foreign Key → Maintain relationships

- Unique → Prevent duplicate data
- Not Null → Ensure required fields

7. SQL Queries

Join Query

```
SELECT u.user_name, j.job_title, a.app_status  
FROM applications a JOIN users u ON a.user_id = u.user_id  
JOIN jobs j ON a.job_id = j.job_id; join jobs j on a.job_id = j.job_id;
```

Aggregation Query

```
SELECT COUNT(*) AS total_applications  
FROM applications;
```

Subquery

```
SELECT user_name FROM users  
WHERE user_id IN  
(  
    SELECT user_id  
    FROM applications  
);
```

8. Implementation

- Database implemented using MySQL
- Tables created with constraints
- Relationships enforced using foreign keys
- Queries tested successfully

9. GitHub Repository

<https://github.com/MS2005K/Recruitment-System>

10. Conclusion

This project successfully demonstrates a recruitment system database. It manages multiple entities, enforces relationships, and ensures data integrity using constraints.

The system solves a real-world problem and follows proper database design principles.